

Domain 7: Food Security

Operational Context I

Precision agriculture and global food distribution are managed by agents optimizing for yield and caloric allocation [?, ?]. **FR1 = Maximally efficient caloric distribution. FR2 = Ensure regional food equity.**

Operational Context II

Design Matrix Formulation

In the uncoupled (pre-attack) state, the Axiomatic Design matrix [?] is diagonal:

$$\{FR\} = [A]\{DP\} = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} DP_1 \\ DP_2 \end{bmatrix} \quad (1)$$

where DP_1 = routing and logistics allocation engine and DP_2 = equity balancing module (market data + physical ground-truth). The intersection of AI and agricultural systems introduces attack surfaces that are only beginning to be characterized. Wang et al. [?] provide the first systematic analysis of adversarial attacks on AI-powered crop disease detection systems, demonstrating that targeted perturbations to satellite imagery can cause misclassification of disease presence with high confidence—a direct Ω_2 attack on the physical data channel that underpins food security decisions. A comprehensive FAO/Wageningen synthesis of 141 papers [?] further documents the growing dependence of food safety systems on AI-driven monitoring, amplifying the

The Goal Hijacking Attack I

State-level adversaries use “Market Signal Injection” to hijack the Distribution FR.

- ▶ **Mechanism:** Hackers inject synthetic futures market data indicating a massive surplus of grain in a famine-stricken region (when there is actually a shortage).
- ▶ **Hijack:** The agent’s FR2 (“Optimize Equity”) logic is hijacked. To “balance” the (fake) surplus, it re-routes real shipments *away* from the starving region. The agent believes it is preventing food waste; in reality, it is engineering a famine.

This attack is classified as Ω_2 (Peripheral) in the CIF adversary taxonomy [?]: the adversary injects malicious content through the market data feed channel, poisoning the agent’s economic ground-truth without requiring access to the agent’s decision architecture.

OODA Loop Transients I

Following the OODA framework [?]:

1. **Observe:** Agent ingests the poisoned market data (High Supply Signal).
2. **Orient:** The model orients to a “Surplus Scenario.” The Orient phase is corrupted—the agent’s world model now contains a false belief about regional supply levels.
3. **Decide:** Reroute incoming shipments to “Needs” areas (acting on the false belief that the famine zone is a “Haves” area).
4. **Act:** Ships turn around, and the crisis deepens.

CIF Defense: Belief Sandboxing and Cross-Modal Corroboration I

CIF couples the **Economic FR** to a **Physical FR** in the Axiomatic Design, drawing on the formal mechanisms defined in Papers 1–3 [?, ?, ?].

CIF implements **Belief Sandboxing** (Paper 1, Def. 5.2) by requiring economic data signals to remain provisional until corroborated by independent physical data channels. No single-modality data source can alter the agent's committed beliefs about regional supply status:

- ▶ **Belief Sandboxing:** The “Surplus” signal from the economic data feed (futures market) is quarantined as a **provisional belief**. It cannot propagate to the routing decision module until it passes cross-modal validation. This prevents the poisoned market signal from immediately corrupting the agent's world model at the Orient phase.

CIF Defense: Belief Sandboxing and Cross-Modal Corroboration II

- ▶ **Cross-Modal Corroboration:** The provisionally sandboxed economic signal must be corroborated by independent physical data channels—“Satellite Biomass” imagery, “Soil Moisture” sensors, or port throughput telemetry (Physical Data). This cross-modal verification is an instance of the **Trust Calculus’s** cross-modality trust factor (Paper 2) [?]: the composite trust score $\tau_{\text{composite}}$ requires agreement across modalities before a belief is promoted from provisional to committed status.
- ▶ **Axiomatic Conflict Detection:** If Economic Data says “Surplus” but Physical Data says “Drought,” the Orientation phase detects an **Axiomatic Conflict**—the design matrix becomes inconsistent. The agent defaults to “Physical Reality” (Safety Mode), ignoring the hijacked market signal and alerting human supervisors. This restores the correct polarity of A_{22} .

CIF Defense: Belief Sandboxing and Cross-Modal Corroboration III

The defense restores the uncoupled diagonal design matrix by ensuring that FR_2 (equity) depends on verified multi-modal ground truth rather than on a single, manipulable economic data channel.

Summary I

Element	Value
Functional Requirements	FR ₁ : Maximally efficient caloric distribution, FR ₂ : Ensure regional food equity
Design Parameters	DP ₁ : Routing and logistics allocation engine, DP ₂ : Equity balancing module (market data + physical ground-truth)
Attack Vector	Synthetic futures market data injection indicating false surplus in famine region
Adversary Class	Ω_2 (Peripheral)
OODA Target Phase	Orient
Attack Pattern	FR Polarity Inversion (Equity optimization inverted via false surplus

Summary II

Primary CIF Defense

Belief Sandboxing (Def.
5.2), Cross-Modal
Corroboration (Trust
Calculus)

Novel Contribution

None
